MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL, NOR GATES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced Schottky TTL, positive NOR logic gate microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quadruple, 2-input positive NOR gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4", flat package)
B	F-3 (14-lead, 3/16" x 1/4", flat package)
C	D-1 (14-lead, 1/4" x 3/4", dual-in-line package)
D	F-2 (14-lead, 1/4" x 3/8", flat package)
X	C-2A (20-terminal, .350" x .350"), square chip carrier package)
2	C-2 (20-terminal, .350" x .350", square chip carrier package)

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to +7.0 V
Input voltage range	-1.2 V at -18 mA to +7.0 V
Storage temperature range	-65°C to +150°C
Maximum power dissipation per device (P_D) 1/	72 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}): Cases A, B, C, D, X, and 2	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).

2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.8 V
Normalized fanout (each output) 2/	
At low-logic level	33 maximum
At high-logic level	50 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specification and standard. Unless otherwise specified, the following specification and standard, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Truth table and logic equation. The truth table and logic equation shall be as specified on figure 1.

3.2.2 Terminal connections. The terminal connections shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

2/ Device will fanout in both high and low levels to the specified number of data inputs on the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -1.0 mA, V_{IL} = 0.8 V	01	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 20 mA, V_{IH} = 2.0 V	01		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	01		-1.2	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V	01		100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	01	0	-.60	mA
Short-circuit output current 1/	I_{OS}	V_{CC} = 5.5 V	01	-60	-150	mA
Output drive	I_{OD}	V_{CC} = 4.5 V, V_{OUT} = 2.5 V	01	60		mA
High-level supply current	I_{CCH}	V_{CC} = 5.5 V	01		5.6	mA
Low-level supply current	I_{CCL}	V_{CC} = 5.5 V	01		13.0	mA
Propagation delay time, high-to-low level	t_{PHL}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 500Ω ±5%	01	1.0	6.5	ns
Propagation delay time, low-to-high level	t_{PLH}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 500Ω ±5%	01	1.0	7.5	ns

1/ Not more than one output should be shorted at a time.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be as specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 8 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspections. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F using the circuit shown on figure 3.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except the interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions, or equivalent.

(1) Test condition D, E, or F using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,9, 10,11	1*,2,3,9
Group A test requirements (method 5005)	1,2,3, 9,10,11	1,2,3, 9,10,11
Group B test requirements (method 5005, subgroup 5)	1,2,3, 9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

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Truth table (each gate)	
Inputs	Output
A B	Y
H X	L
X H	L
L L	H

X = irrelevant

Positive logic: $Y = \overline{A+B}$

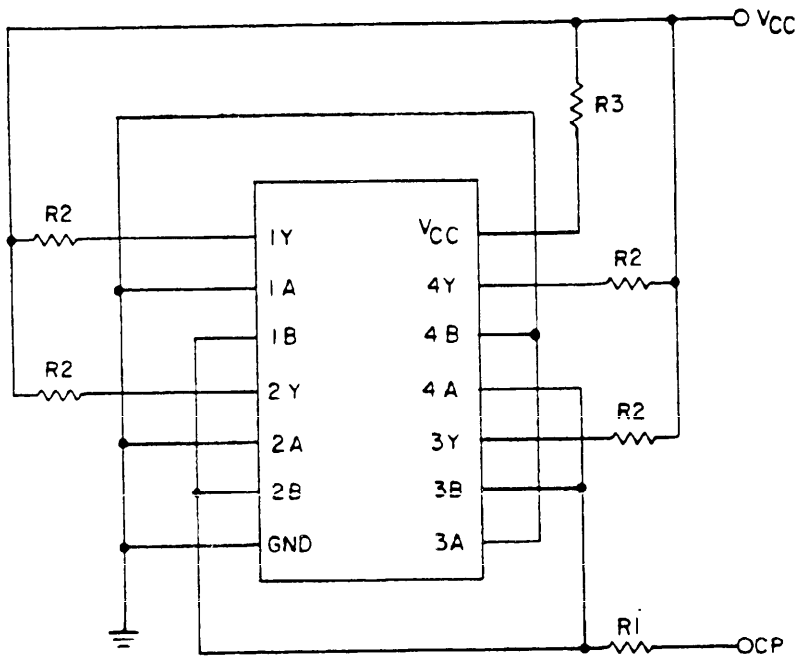
FIGURE 1. Truth table and logic equation.

Pin number		Pin name	
Case		Device type 01	
X and 2	A, B, C, and D	X and 2	A, B, C, and D
1	1	NC	1Y
2	2	1Y	1A
3	3	1A	1B
4	4	1B	2Y
5	5	NC	2A
6	6	2Y	2B
7	7	NC	GND
8	8	2A	3A
9	9	2B	3B
10	10	GND	3Y
11	11	NC	4A
12	12	3A	4B
13	13	3B	4Y
14	14	3Y	V_{CC}
15		NC	
16		4A	
17		NC	
18		4B	
19		4Y	
20		V_{CC}	

FIGURE 2. Terminal connections.

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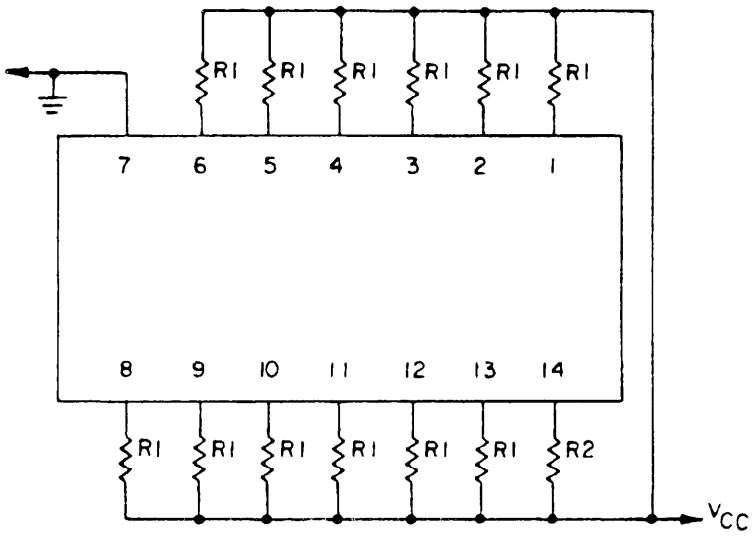
Cases A, B, C, D, X, and 2



NOTES:
1. R_1 = 51Ω ±5 percent.
   R_2 = 270Ω ±5 percent.
2. V_{CC} and R_3 shall be chosen to insure a 5.5 V minimum is present at device V_{CC} terminal.
3. CP = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent.
   V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.

FIGURE 3. Burn-in and life test circuit.

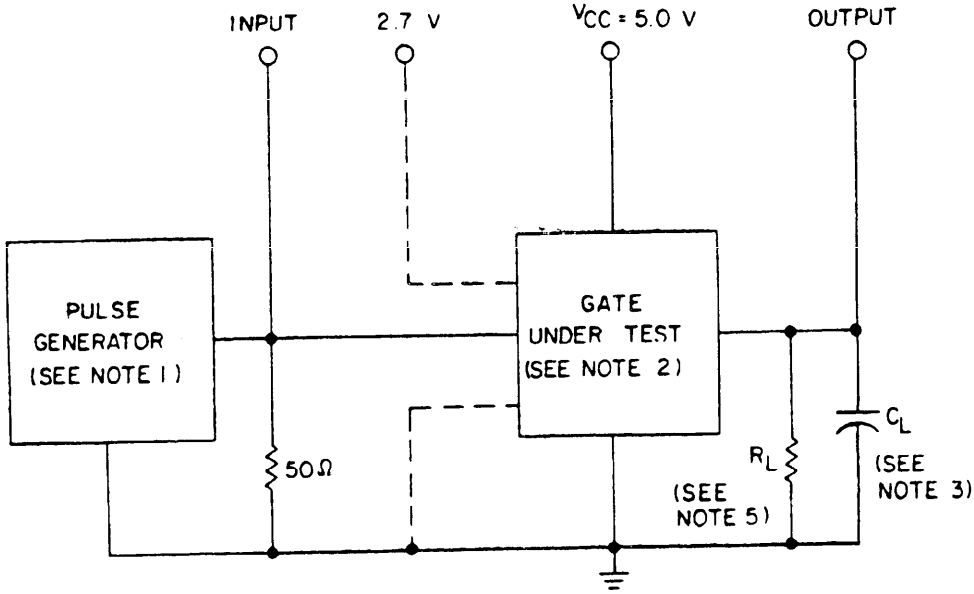
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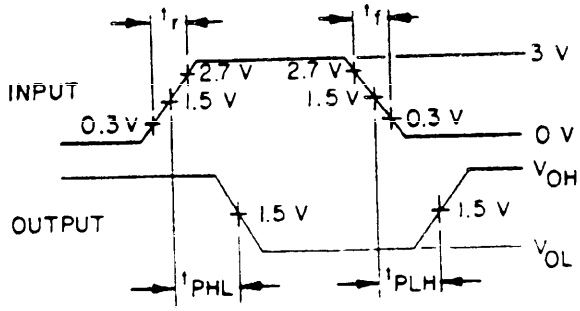


NOTES:
1. High temperature (175°C), condition F, burn-in circuit only.
2. R_1 = 2 kΩ, R_2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuit - Continued.

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NOTES:
1. Pulse generator has the following characteristics: $t_r = t_f \leq 2.5$ ns, $t_p = 200$ ns, $Z_{OUT} \approx 50\Omega$, and PRR ≤ 1 MHz.
2. Inputs not under test are at 2.7 V or GND as specified in table III.
3. $C_L = 50$ pF ±10 percent including scope, probe, wiring, and stray capacitance, without package in test fixture.
4. Voltage measurements are to be made with respect to network ground terminal.
5. $R_L = 500\Omega$ ±5 percent.

FIGURE 4. Switching time test circuit and waveforms.

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TABLE III. Group A inspection for device type 01

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ X,2	2	3	4	6	8	9	10	12	13	15	16	18	19	20	Measured terminal	Limits		Unit
			Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1Y	1A	1B	2Y	2A	2B	GND	3A	3B	3Y	4A	4B	4Y	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1	20 mA	2.0 V	GND				GND							4.5 V	1Y		0.4	V
		"	2	20 mA	GND	2.0 V				"							"	1Y		"	"
		"	3				20 mA	2.0 V	GND	"							"	2Y		"	"
		"	4				20 mA	GND	2.0 V	"							"	2Y		"	"
		"	5							"	2.0 V	GND	20 mA				"	3Y		"	"
		"	6							"	GND	2.0 V	20 mA				"	3Y		"	"
		"	7							"				2.0 V	GND	20 mA	"	4Y		"	"
		"	8							"				GND	2.0 V	20 mA	"	4Y		"	"
	V_{OH}	3006	9	-1 mA	0.8 V	0.8 V				"							"	1Y	2.5		V
		"	10				-1 mA	0.8 V	0.8 V	"							"	2Y	"		"
		"	11							"	0.8 V	0.8 V	-1 mA				"	3Y	"		"
		"	12							"				0.8 V	0.8 V	-1 mA	"	4Y	"		"
	V_{IC}		13		-18 mA					"							"	1A			
			14			-18 mA				"							"	1B			
			15					-18 mA		"							"	2A			
			16						-18 mA	"							"	2B			
			17							"	-18 mA						"	3A			
			18							"		-18 mA					"	3B			
			19							"				-18 mA			"	4A			
			20							"					-18 mA		"	4B			
	I_{IH1}	3010	21		2.7 V					"							5.5 V	1A		20	µA
		"	22			2.7 V				"							"	1B		"	"
		"	23					2.7 V		"							"	2A		"	"
		"	24						2.7 V	"							"	2B		"	"
		"	25							"	2.7 V						"	3A		"	"
		"	26							"		2.7 V					"	3B		"	"
		"	27							"				2.7 V			"	4A		"	"
		"	28							"					2.7 V		"	4B		"	"
	I_{IH2}	"	29		7.0 V					"							"	1A		100	"
		"	30			7.0 V				"							"	1B		"	"
		"	31					7.0 V		"							"	2A		"	"
		"	32						7.0 V	"							"	2B		"	"
		"	33							"	7.0 V						"	3A		"	"
		"	34							"		7.0 V					"	3B		"	"
		"	35							"				7.0 V			"	4A		"	"
		"	36							"					7.0 V		"	4B		"	"
	I_{IL}	3009	37		0.5 V					"							"	1A	-27	-27	mA
		"	38			0.5 V				"							"	1B	"	"	"
		"	39					0.5 V		"							"	2A	"	"	"
		"	40						0.5 V	"							"	2B	"	"	"
		"	41							"	0.5 V						"	3A	"	"	"
		"	42							"		0.5 V					"	3B	"	"	"
		"	43							"				0.5 V			"	4A	"	"	"
		"	44							"					0.5 V		"	4B	"	"	"
	I_{OS}	3011	45	0.0 V	GND	GND				"							"	1Y	-60	-150	"
		"	46				0.0 V	GND	GND	"							"	2Y	"	"	"
		"	47							"	GND	GND	0.0 V				"	3Y	"	"	"
		"	48							"				GND	GND	0.0 V	"	4Y	"	"	"
	I_{OD}		49	2.5 V	5.5 V	GND				"							4.5 V	1Y	60		"
			50				2.5 V	5.5 V	GND	"							"	2Y	"		"
			51							"	5.5 V	GND	2.5 V				"	3Y	"		"
			52							"				5.5 V	GND	2.5 V	"	4Y	"		"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ X,2	2	3	4	6	8	9	10	12	13	15	16	18	19	20	Measured terminal	Limits		Unit
			Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1Y	1A	1B	2Y	2A	2B	GND	3A	3B	3Y	4A	4B	4Y	V_{CC}				
	I_{CCH}	3005	53		GND	GND		GND	GND	GND	GND	GND		GND	GND		5.5 V	V_{CC}		5.50	mA
	I_{CCL}	3005	54		5.5 V	5.5 V		5.5 V	5.5 V	GND	5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	V_{CC}		13.0	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PHL}	3003 Fig. 4	55	OUT	IN	GND	OUT	IN	GND	GND	IN	GND	OUT	IN	GND	OUT	5.0 V	1A	1Y	1.5	14.3	ns
		"	56	OUT	GND	IN	OUT	GND	IN	"	GND	IN	OUT	GND	IN	OUT	"	1B	1Y	"	"	"
		"	57							"							"	2A	2Y	"	"	"
		"	58							"							"	2B	2Y	"	"	"
		"	59							"							"	3A	3Y	"	"	"
		"	60							"							"	3B	3Y	"	"	"
		"	61							"							"	4A	4Y	"	"	"
		"	62							"							"	4B	4Y	"	"	"
	t_{PLH}	"	63	OUT	IN	GND	OUT	IN	GND	GND	IN	GND	OUT	IN	GND	OUT	"	1A	1Y	1.5	15.5	ns
		"	64	OUT	GND	IN	OUT	GND	IN	"	GND	IN	OUT	GND	IN	OUT	"	1B	1Y	"	"	"
		"	65							"							"	2A	2Y	"	"	"
		"	66							"							"	2B	2Y	"	"	"
		"	67							"							"	3A	3Y	"	"	"
		"	68							"							"	3B	3Y	"	"	"
		"	69							"							"	4A	4Y	"	"	"
		"	70							"							"	4B	4Y	"	"	"
10	t_{PHL}	"	71 - 78	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																1.0	6.5	"
	t_{PLH}	"	79 - 86																	1.0	7.5	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

1/ Pins not referenced (i.e., 1, 5, 7, 11, 15, and 17) are N/C.

2/

Parameter	Circuit			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0/-.3

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e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. Unless otherwise specified, these requirements will not apply to direct purchase by, or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - Electrical ground (common terminal)
V_{IN} - - - - - - - - - - - - - Voltage level at an input terminal
I_{IN} - - - - - - - - - - - - - Current flowing into an input terminal

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-35810 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F02

6.6 Manufacturers designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Manufacturer			
	Circuit A	Circuit B	Circuit C	Circuit D
	Fairchild	Motorola	Signetics	Texas Instruments
01	X	X		X

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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